

Parameter

The SB_BARCODE_DRV primitive contains the following parameter and their default values:

Parameter CURRENT_MODE = "0b0";

Parameter values:

"0b0" = Full Current Mode (Default).

"0b1" = Half Current Mode.

Parameter BARCODE_CURRENT = "0b0000";

Parameter values:

"0b0000" = 0mA. //This is the setting to tristate the BARCODEoutput to allow it to be used
// as GPIO (SB_IO_OD)

"0b0001" = 16.6mA for Full Mode; 8.3mA for Half Mode,

"0b0011" = 33.3mA for Full Mode; 16.6mA for Half Mode,

"0b1001" = 66.6mA for Full Mode; 33.3mA for Half Mode,

"0b1010" = 83.3mA for Full Mode; 41.6mA for Half Mode,

"0b0111" = 50mA for Full Mode; 25mA for Half Mode,

"0b1111" = 100mA for Full Mode; 50mA for Half Mode

Verilog Instantiation

```
SB_BARCODE_DRV BARCODEDRVinst (  
    .CURREN(ENABLE_CURRENT),  
    .BARCODEEN(ENABLE_BARCODEDRV),  
    .BARCODEPWM(BARCODE_PWMINPUT),  
    .BARCODE(BARCODEOUT)  
);  
  
defparam BARCODEDRVinst.CURRENT_MODE = "0b0";  
defparam BARCODEDRVinst.BARCODE_CURRENT = "0b1111";
```

SB_IR500_DRV

SB_IR500_DRV primitive is the IR driver which contains a two dedicated open drain I/O pin for IRLED1, IRLED2 outputs. The IRLED outputs are bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_IR500_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

When the IR4500_CURRENT parameter is set to "0b00000000", then SB_IO_OD can be used to drive the package pin.



SB_IR500_DRV DRC Rule

This primitive cannot be instantiated along with SB_BARCODE_DRV or SB_IR400_DRV instance.

Ports

SB_IR500_DRV Ports		
Signal Name	Direction	Description
CURREN	Input	Enable the mixed signal control block to supply reference current to the IR drivers. When it is not enabled (CURREN=0), no current is supplied, and the IR drivers are powered down. Enabling the mixed signal control block takes 100us to reach a stable reference current value.
IRLEDEN	Input	Enable the SB_IR400_DRV primitive. Active High.
IRPWM	Input	PWM Input data to drive IRLED pin.
IRLED1	Output	IR LED output 1.
IRLED2	Output	IR LED output 2.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:

Input CURREN : Logic "0"

Input IRLEDEN : Logic "0"

Input IRPWM : Logic "0"

Parameter

The SB_IR500_DRV primitive contains the following parameter and their default values:

Parameter CURRENT_MODE = "0b0";

Parameter values:

"0b0"; = Full Current Mode (Default).

"0b1"; = Half Current Mode.

Parameter IR500_CURRENT = "0b000000000000";

Parameter values:

"0b000000000000"; = 0mA. // This is the setting to tristate the BARCODE output to allow it to
// be used as GPIO (SB_IO_OD).

"0b000000000111"; = 50mA for Full Mode; 25mA for Half Mode.

"0b000000001111"; = 100mA for Full Mode; 50mA for Half Mode.

"0b000000011111"; = 150mA for Full Mode; 75mA for Half Mode.

"0b000000111111"; = 200mA for Full Mode; 100mA for Half Mode.

"0b000001111111"; = 250mA for Full Mode; 125mA for Half Mode.

"0b000011111111"; = 300mA for Full Mode; 150mA for Half Mode.

"0b000111111111"; = 350mA for Full Mode; 175mA for Half Mode.

"0b001111111111"; = 400mA for Full Mode; 200mA for Half Mode.

"0b011111111111"; = 450mA for Full Mode; 225mA for Half Mode.

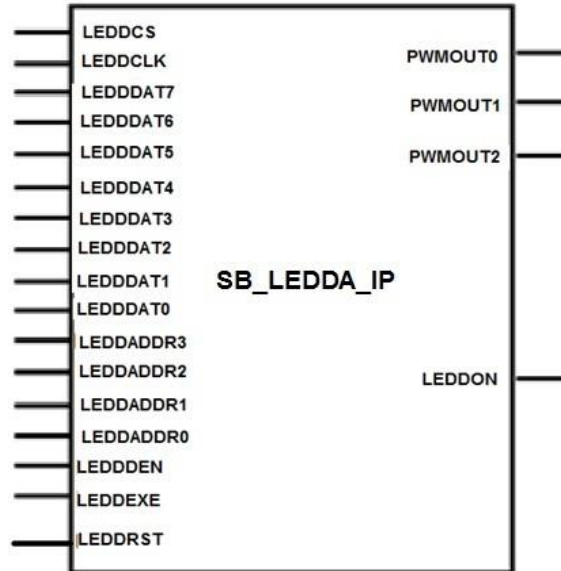
"0b111111111111"; = 500mA for Full Mode; 250mA for Half Mode.

Verilog Instantiation

```
SB_IR500_DRV IRDRVinst (  
    .CURREN(ENABLE_CURRENT),  
    .IRLEDEN(ENABLE_IRDRV),  
    .IRPWM(IR_PWMINPUT),  
    .IRLED(IR_LEDOUT)  
);  
  
defparam IRDRVinst.CURRENT_MODE = "0b0";  
defparam IRDRVinst.IR500_CURRENT = "0b1111111111";
```

SB_LEDDA_IP

SB_LEDDA_IP primitive generates the RGB PWM outputs for the RGB LED drivers. The IP contains registers that are programmed in by the SCI bus interface signals.



Ports

SB_LEDDA_IP Ports		
Signal Name	Direction	Description
LEDDCS	Input	CS to write LEDD IP registers
LEDDCLK	Input	Clock to write LEDD IP registers
LEDDDAT7	Input	bit 7 data to write into the LEDD IP registers
LEDDDAT6	Input	bit 6 data to write into the LEDD IP registers
LEDDDAT5	Input	bit 5 data to write into the LEDD IP registers
LEDDDAT4	Input	bit 4 data to write into the LEDD IP registers
LEDDDAT3	Input	bit 3 data to write into the LEDD IP registers
LEDDDAT2	Input	bit 2 data to write into the LEDD IP registers
LEDDDAT1	Input	bit 1 data to write into the LEDD IP registers
LEDDDAT0	Input	bit 0 data to write into the LEDD IP registers
LEDDADDR3	Input	LEDD IP register address bit 3
LEDDADDR2	Input	LEDD IP register address bit 2
LEDDADDR1	Input	LEDD IP register address bit 1
LEDDADDR0	Input	LEDD IP register address bit 0
LEDDDEN	Input	Data enable input to indicate data and address are stable.
LEDDEXE	Input	Enable the IP to run the blinking sequence. When it is LOW, the sequence stops at the nearest OFF state.
LEDDRST	Input	Device level reset signal to reset all internal registers during the device configuration. This port is not accessible to user signals.
PWMOUT0	Output	PWM output 0
PWMOUT1	Output	PWM output 1
PWMOUT2	Output	PWM output 2

LEDDON	Output	Indicating the LED is on
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Default Signal Values

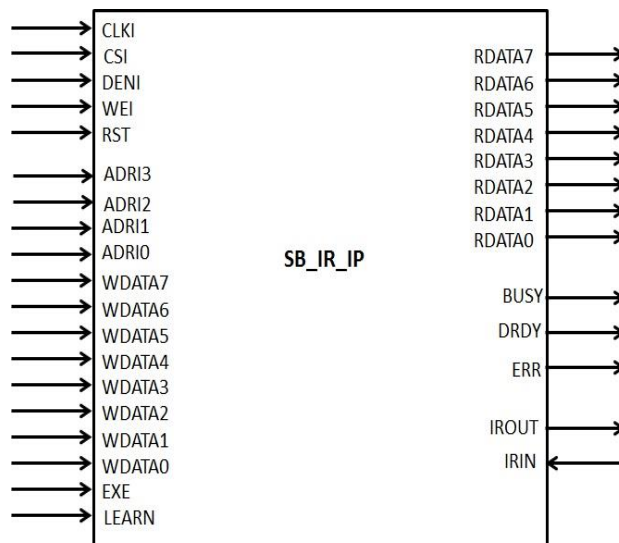
The iCEcube2 software assigns the logic “0” value to all unconnected input ports.

Verilog Instantiation

```
SB_LEDDA_IP PWMgen_inst (
    .LEDDCS(led_cs),
    .LEDDCLK(led_clk),
    .LEDDDAT7(led_ip_data[7]),
    .LEDDDAT6(led_ip_data[6]),
    .LEDDDAT5(led_ip_data[5]),
    .LEDDDAT4(led_ip_data[4]),
    .LEDDDAT3(led_ip_data[3]),
    .LEDDDAT2(led_ip_data[2]),
    .LEDDDAT1(led_ip_data[1]),
    .LEDDDAT0(led_ip_data[0]),
    .LEDDADDR3(led_ip_addr[3]),
    .LEDDADDR2(led_ip_addr[2]),
    .LEDDADDR1(led_ip_addr[1]),
    .LEDDADDR0(led_ip_addr[0]),
    .LEDDDEN(led_ip_den),
    .LEDDEXE(led_ip_exe),
    .LEDDRST(led_ip_rst),
    .PWMOUT0(LED0),
    .PWMOUT1(LED1),
    .PWMOUT2(LED2),
    .LEDDON(led_on)
);
```

SB_IR_IP

SB_IR_IP primitive is the IR transceiver module. It generates or receives the modulated pulse for the IR driver primitives.



Ports

SB IR IP Ports		
Signal Name	Direction	Description
CLKI	Input	Clock input for IR IP
CSI	Input	Select Signal. Active High to activate the IP. This usually connects to the output of the decoding logic from MSB of the address bus
DENI	Input	Data Enable. When asserted, indicates that the data and address on the IR Transceiver Control Bus are stabilized and ready to be captured.
WEI	Input	Data Write Enable. Asserted during WRITE and de-asserted during READ cycle.
ADRI3	Input	Control Register Address Bit 3
ADRI2	Input	Control Register Address Bit 2
ADRI1	Input	Control Register Address Bit 1
ADRI0	Input	Control Register Address Bit 0
WDATA7	Input	Write Data Input Bit 7
WDATA6	Input	Write Data Input Bit 6
WDATA5	Input	Write Data Input Bit 5
WDATA4	Input	Write Data Input Bit 4
WDATA3	Input	Write Data Input Bit 3
WDATA2	Input	Write Data Input Bit 2
WDATA1	Input	Write Data Input Bit 1
WDATA0	Input	Write Data Input Bit 0
RDATA7	Output	Read Data Output Bit 7
RDATA6	Output	Read Data Output Bit 6
RDATA5	Output	Read Data Output Bit 5
RDATA4	Output	Read Data Output Bit 4
RDATA3	Output	Read Data Output Bit 3
RDATA2	Output	Read Data Output Bit 2
RDATA1	Output	Read Data Output Bit 1
RDATA0	Output	Read Data Output Bit 0
EXE	Input	Execute. when asserted, starts the IR Transceiver Hard IP to transmit or receive IR data
LEARN	Input	Learning Mode control. When asserted the IR Transceiver is in learning mode. The IR Transceiver will receive data instead of transmit data.
BUSY	Output	Busy status output
DRDY	Output	Data Buffer Ready status output
ERR	Output	Data Error status
RST	Input	Device level reset signal to reset all internal registers and IROUT signal to OFF state during the device configuration. This port is not accessible to user signals.
IRIN	Input	Modulated ON/OFF pulse from IR sensor.
IROUT	Output	Modulated ON/OFF pulse for IR Transmit.

Default Signal Values

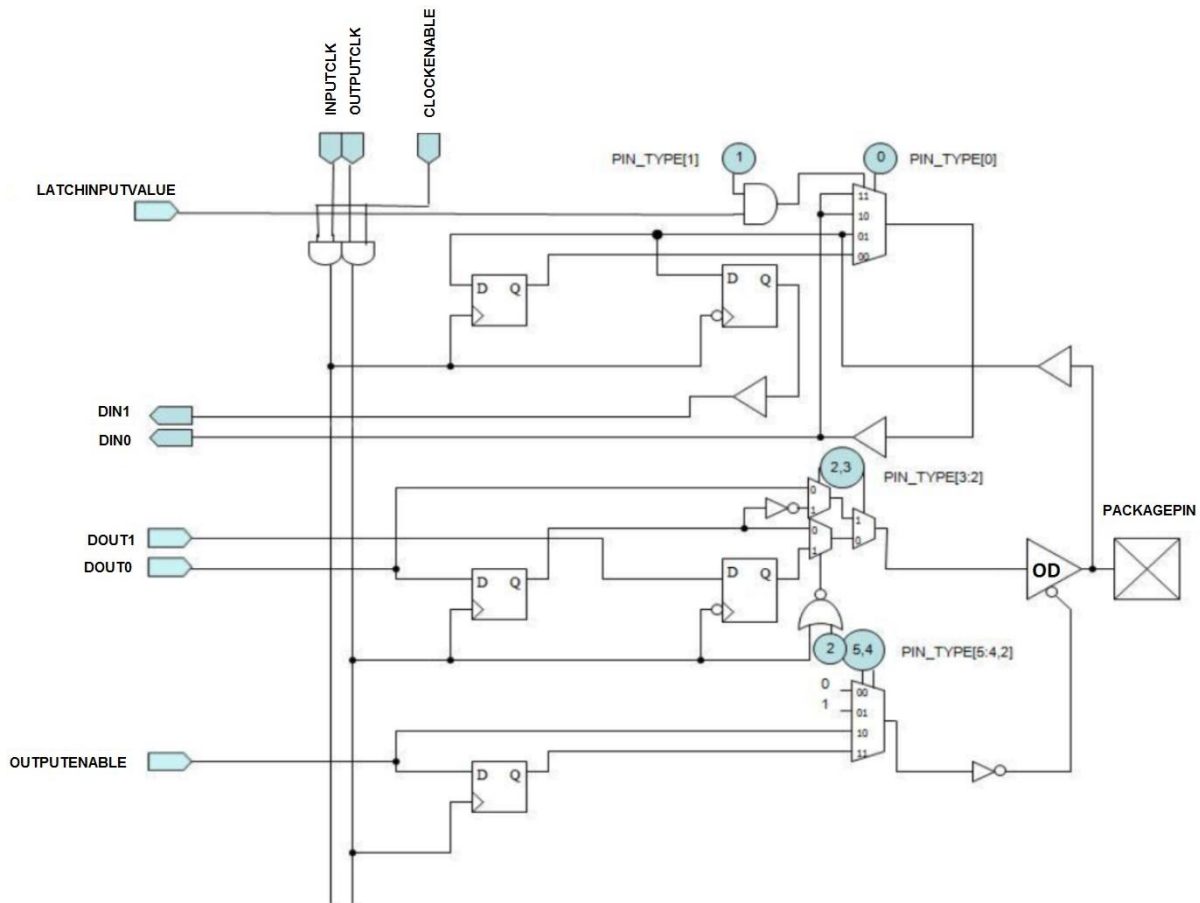
The iCEcube2 software assigns the logic “0” value to all unconnected input ports.

Verilog Instantiation

```
SB_IR_IP IRIP_inst (  
    .CLKI(sysclk_i),  
    .CSI(csi_i),  
    .DENI(deni_i),  
    .WEI(wei_i),  
    .ADRI3(addr_i[3]),  
    .ADRI2(addr_i[2]),  
    .ADRI1(addr_i[1]),  
    .ADRI0(addr_i[0]),  
    .WDATA7(wdata_i[7]),  
    .WDATA6(wdata_i[6]),  
    .WDATA5(wdata_i[5]),  
    .WDATA4(wdata_i[4]),  
    .WDATA3(wdata_i[3]),  
    .WDATA2(wdata_i[2]),  
    .WDATA1(wdata_i[1]),  
    .WDATA0(wdata_i[0]),  
    .RDATA7(rdata_o[7]),  
    .RDATA6(rdata_o[6]),  
    .RDATA5(rdata_o[5]),  
    .RDATA4(rdata_o[4]),  
    .RDATA3(rdata_o[3]),  
    .RDATA2(rdata_o[2]),  
    .RDATA1(rdata_o[1]),  
    .RDATA0(rdata_o[0]),  
    .EXE(exe_i),  
    .LEARN(learn_i),  
    .BUSY(busy_o),  
    .DRDY(drdy_o),  
    .ERR(err_o),  
    .RST(rst_i),  
    .IRIN(irin_i),  
    .IROUT(irpulse_w)  
);
```

SB_IO_OD

The SB_IO_OD is the open drain IO primitive. When the Tristate output is enabled, the IO pulls down the package pin signal to zero. The following figure and Verilog template illustrate the complete user accessible logic diagram, and its Verilog instantiation.



Ports

SB_IO_OD Ports		
Signal Name	Direction	Description
PACKAGEPIN	Bidirectional	Bidirectional IO pin.
LATCHINPUTVALUE	Input	Latches/Holds the input data.
CLOCKENABLE	Input	Clock enable signal.
INPUTCLK	Input	Clock for the input registers.
OUTPUTCLK	Input	Clock for the output registers.
OUTPUTENABLE	Input	Enable Tristate output. Active high.
DOUT0	Input	Data to package pin.
DOUT1	Input	Data to package pin.
DIN0	Output	Data from package pin.
DIN1	Output	Data from package pin.

Default Signal Values

The iCEcube2 software assigns the logic “0” value to all unconnected input ports except for CLOCKENABLE.

Note that explicitly connecting logic “1” value to port CLOCKENABLE will result in a non-optimal implementation, since extra LUT will be used to generate the Logic “1”. If the user’s intention is to keep the Input and Output registers always enabled, it is recommended that port CLOCKENABLE to be left unconnected.

Parameter Values

```
Parameter PIN_TYPE = 6'b000000;  
                                // See Input and Output Pin Function Tables in SB_IO.  
                                // Default value of PIN_TYPE = 6'b000000  
Parameter NEG_TRIGGER = 1'b0;  
                                // Specify the polarity of all FFs in the I/O to be falling edge when  
                                NEG_TRIGGER = 1. Default is 1'b0, rising edge.
```

Input and Output Pin Function Tables

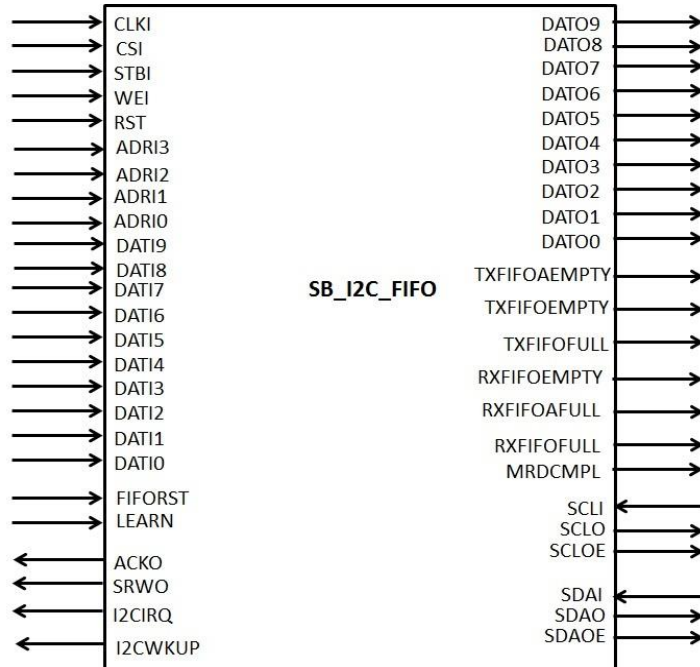
Refer SB_IO Input and Output Pin Functional Table for the PIN_TYPE settings. Some of the output pin configurations are not applicable for SB_IO_OD primitive.

Verilog Instantiation

```
SB_IO_OD    OpenDrainInst0  
(  
  .PACKAGEPIN (PackagePin),           // User's Pin signal name  
  .LATCHINPUTVALUE (latchinputvalue), // Latches/holds the Input value  
  .CLOCKENABLE (clockenable),          // Clock Enable common to input and  
                                        // output clock  
  .INPUTCLK (inputclk),                // Clock for the input registers  
  .OUTPUTCLK (outputclk),              // Clock for the output registers  
  .OUTPUTENABLE (outputenable),        // Output Pin Tristate/Enable  
                                        // control  
  .DOUT0 (dout0),                      // Data 0 - out to Pin/Rising clk  
                                        // edge  
  .DOUT1 (dout1),                      // Data 1 - out to Pin/Falling clk  
                                        // edge  
  .DIN0 (din0),                        // Data 0 - Pin input/Rising clk  
                                        // edge  
  .DIN1 (din1),                        // Data 1 - Pin input/Falling clk  
                                        // edge  
);  
  
defparam OpenDrainInst0.PIN_TYPE = 6'b000000;  
defparam OpenDrainInst0.NEG_TRIGGER = 1'b0;
```

SB_I2C_FIFO

iCE40UL device supports two I2C hard IP primitives , located at bottom left corner and bottom right corner of the chip.



Ports

SB_I2C_FIFO Ports		
Signal Name	Direction	Description
CLKI	Input	System Clock input
CSI	Input	Select Signal. Active High to activate the IP. This usually connects to the output of the decoding logic from MSB of the address bus
STBI	Input	Strobe Signal
WEI	Input	Write Enable Signal
ADRI3	Input	Control Register Address Bit 3
ADRI2	Input	Control Register Address Bit 2
ADRI1	Input	Control Register Address Bit 1
ADRI0	Input	Control Register Address Bit 0
DATI9	Input	Data Input Bit 9
DATI8	Input	Data Input Bit 8
DATI7	Input	Data Input Bit 7
DATI6	Input	Data Input Bit 6
DATI5	Input	Data Input Bit 5
DATI4	Input	Data Input Bit 4
DATI3	Input	Data Input Bit 3
DATI2	Input	Data Input Bit 2
DATI1	Input	Data Input Bit 1
DATI0	Input	Data Input Bit 0
DATO9	Output	Data Output Bit 9
DATO8	Output	Data Output Bit 8
DATO7	Output	Data Output Bit 7

DATO6	Output	Data Output Bit 6
DATO5	Output	Data Output Bit 5
DATO4	Output	Data Output Bit 4
DATO3	Output	Data Output Bit 3
DATO2	Output	Data Output Bit 2
DATO1	Output	Data Output Bit 1
DATO0	Output	Data Output Bit 0
ACKO	Output	Acknowledgement
I2CIRQ	Output	I2C Interrupt output
I2CWKUP	Output	I2C Wake up from Standby signal
SRWO	Output	Slave RW "1" master receiving / Slave transmitting "0" master transmitting / Slave receiving
SCLI	Input	Serial Clock Input
SCLO	Output	Serial Clock Output
SCLOE	Output	Serial Clock Output Enable. Active High
SDAI	Input	Serial Data Input
SDAO	Output	Serial Data Output
SDAOE	Output	Serial Data Output Enable. Active High
FIFORST ¹	Input	RESET for the FIFO logic
TXFIFOEMPTY ¹	Output	TX FIFO Status Signal
TXFIFOEMPTY ¹	Output	TX FIFO Status Signal
TXFIFOFULL ¹	Output	TX FIFO Status Signal
RXFIFOAFULL ¹	Output	RX FIFO Status Signal
RXFIFOFULL ¹	Output	RX FIFO Status Signal
RXFIFOEMPTY ¹	Output	RX FIFO Status Signal
MRDCMPL ¹	Output	Master Read Complete (only valid for Master Read Mode)

Note ¹: Only available if I2C_FIFO_ENB = ENABLED.

Parameters

I2C Location	Parameters	Parameter Default	Description
Left Side Corner	I2C_SLAVE_ADDR	0b1111100001	Upper Bits <9:2> can be changed through control registers. Lower bits <1:0> are fixed.
Right Side	I2C_SLAVE_ADDR	0b1111100010	Upper Bits <9:2> can be changed through control registers. Lower bits <1:0> are fixed.

Synthesis Attribute

I2C_CLK_DIVIDER

Synthesis attribute "I2C_CLK_DIVIDER" is used by PNR and STA tools for optimization and deriving the appropriate clock frequency at SCLO output with respect to the SBCLKI input clock frequency.

/* synthesis I2C_CLK_DIVIDER= Divide Value */
Divide Value: 0, 1, 2, 3 ... 1023. Default is 0.

SDA_INPUT_DELAYED

SDA_INPUT_DELAYED attribute is used to add 50ns additional delay to the SDAI signal.

/* synthesis SDA_INPUT_DELAYED= value */